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Gender differences and measurement bias in the assessment of adult intelligence: Evidence from the Italian WAIS-IV and WAIS-R standardizations

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ABSTRACT

Research on gender differences in intelligence has led to inconsistent results. Different methods seem to provide different estimates using the Wechsler scales. The present study analyzed WAIS-IV and WAIS-R standardization data. We assessed gender differences in subtest scores, indices, and IQs and used a Multi-Group Confirmatory Factor Analysis (MGCFA) to investigate whether gender differences were due to differences in ability factors or test bias. The larger effect sizes in favor of men were found in the Arithmetic subtest and the Working Memory Index of the WAIS-IV. Likewise, for the WAIS-R, Arithmetic, Information, and Block Design subtests and all IQ scores were the larger effect sizes. The MGCFA supported the metric invariance of both editions. For the WAIS-IV, partial scalar invariance was achieved by releasing intercept equality for Information, Comprehension, and Arithmetic. For the WAIS-R, also Coding and Digit Span intercepts had to be unconstrained. Men outperformed women on the perceptual reasoning ability factor. No gender differences were found for the processing speed factor. Men had higher working memory ability, while no difference emerged for verbal comprehension. However, the latent mean differences could be affected by a lack of scalar invariance. A comparison of observed effect sizes at the subtest level to those expected according to MGCFA revealed that Information, Arithmetic, and Comprehension subtests were gender-biased in both editions. As a whole, gender differences appeared more pronounced in the WAIS-R. Gender differences in perceptual reasoning seem genuine, as does their absence in processing speed. Implications for clinical assessment are discussed.

1. Introduction

Ever since intelligence tests have been introduced, the question of whether there are gender differences in cognitive abilities has always been a controversial matter. Research on gender differences in intelligence has lead to inconsistent results on the presence, magnitude, and direction of these effects. Studies have shown a small average gender difference in general intelligence favoring adolescent and adult males (Irwing, 2012; Irwing & Lynn, 2005; Jackson & Rushton, 2006; Lynn, 1999; Lynn & Irwing, 2004a, 2004b, 2008; Nyborg, 2003; Nyborg, 2005). By contrast, other studies have found a women's advantage in general intelligence (Keith, Reynolds, Patel, & Ridley, 2008; Reynolds, Keith, Ridley, & Patel, 2008). However, there is also evidence of no or negligible gender differences in intelligence (e.g., Aluja-Fabregat, Colom, Abad, & Juan-Espinosa, 2000; Colom, García, Juan-Espinosa, & Abad, 2002; Colom, Juan-Espinosa, Abad, & García, 2000; Pezzuti & Orsini, 2016; Saggino et al., 2014; Tommasi et al., 2015).

Gender differences in intelligence have been interpreted concerning both substantive and methodological issues. These include developmental effects (Lynn, 1999), brain size (Ankney, 1992; Lynn, 1994; McDaniel, 2005; Nyborg, 2003; Rushton & Ankney, 2009), the operationalization of general and broad abilities in different tests (Colom & Garcia-Lopez, 2002; Lynn, 1999), and psychometric approaches in data analyses (Keith et al., 2008; Steinmayr, Beauducel, & Spinath, 2010). Among methodological considerations, the representativeness of male and female samples has attracted considerable attention (Dykiert, Gale, & Deary, 2009; Hunt & Madhyastha, 2008). Most of the researches that have studied the effects of gender used normative samples of intelligence scales collected in various countries, with no insurance of full representativeness regarding gender, age, and level of education. Also,

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not all the eligible respondents eventually agreed to participate. Not always, these sampling and volunteer biases are taken into account in standardization samples.

Intelligence is a complex system of cognitive abilities (Rindermann, 2007). Therefore it is reasonable to expect an articulate pattern of cognitive performances between males and females. Considering broad cognitive abilities, such as verbal, numerical, or figural intelligence, Terman (1916) and Wechsler (1944) already noted a male advantage in complex numerical tasks and requiring figural abilities, while women showed a higher score in some verbal tasks and measures of perceptual speed. In general, research has shown that men typically outperform women in visual-spatial tasks, whereas women outperform men in tests of processing speed (e.g., Keith et al., 2008; Longman, Saklofske, & Fung, 2007). Nevertheless, gender differences in broad areas of cognitive abilities have evolved. Indeed, there is recent evidence that males and females perform equally on mathematics assessments, the gender difference in verbal skills is small, while there is still a moderate male advantage in tasks requiring spatial abilities (Hyde, 2016).

Studies of gender differences that took advantage of the Wechsler scales compared means of subtests and indices (e.g., Daseking, Petermann, & Waldmann, 2017; Longman et al., 2007; Lynn, 1998), examined latent mean differences in Multigroup Confirmatory Factor Analysis (MGCFA) (e.g., Colom & Garcia-Lopez, 2002; Dolan et al., 2006; Tommasi et al., 2015; Van der Sluis et al., 2006, 2008) or used both methods (e.g., Irwing, 2012; Saggino et al., 2014). The results are generally not consistent also depending on the method used.

Studies that examined the gender differences using subtests and derived indices have shown an advantage of men in IQ, Verbal Comprehension (VC), Perceptual Reasoning (PR), and Working Memory (WM) indices (Daseking et al., 2017; Irwing, 2012; Longman et al., 2007). Instead, women scored significantly higher on Processing Speed (PS) index. The studies that estimated the latent mean differences yielded different results. For example, while both Dolan et al. (2006) and Van der Sluis et al. (2008) did not find any difference in the general ability factor, Irwing (2012) found it. The latent mean difference never turned statistically significant for the VC factor in all studies (Dolan et al., 2006; Irwing, 2012; Van der Sluis et al., 2008). Different from Irwing (2012), other studies showed that the latent mean difference for PR and WM was higher for men than for women (Dolan et al., 2006; Van der Sluis et al., 2008). As for the PS factor, the advantage of women was confirmed at the level of latent mean differences but only in two studies (Dolan et al., 2006; Irwing, 2012; Van der Sluis et al., 2008). Interestingly, Irwing (2012) also analyzed the observed mean differences. He found different results between the two analyses that he interpreted as due to an alleged difference in the g factor. This interpretation has some face validity but is inconsistent with the women's advantage in the processing speed factor. Indeed, a presumed gender difference in average general intelligence cannot explain why only PR and WM showed a male advantage while PS a female one.

Gender differences in broad abilities could be best explained as a mixture of subtest specific differences and differences in various first-order factors. All the comparisons carried out using subtests and indices could not separate the different components (at the level of subtests and latent factors) that might contribute to generating differences related to the gender one belongs to. Following Wicherts and Dolan (2010), it is vital to assess whether observed gender effects reflect a genuine difference in the latent abilities or are because of other factors than abilities (e.g., test bias or other subtest specific processes). If women and men differ on specific subtests and these differences reflect genuine differences in broad ability factors, the effect sizes at the subtest level should match the corresponding factor loadings. In other words, a gender difference equal to one unit of measurement at each subtest level (i.e., the observed effect size) should be proportional to the strength with which that subtest binds to the corresponding factor (i.e., the factor loading). If the effect size is disproportionate to the factor loadings, there should be something subtest-specific (but unrelated to the latent ability) that produces an unexpected gender difference. Wicherts and Dolan (2010) have provided an enlightening description of a method by which one can rescale the factor loadings to compare them to the observed effect size of specific subtests. This method, applied to the study of gender differences in the subtests of the WAIS-IV (and of the WAIS-R, for comparison), could provide an insight on which subtest could be more gender-biased or fair.

When the gender effect size of a subtest does not match with the factor loading, there might be a gender difference in the intercept of that subtest that must be freed across groups in an MGCFA to achieve an acceptable fit. The more unconstrained intercepts are in the model, the more the latent mean of the corresponding ability factor is biased. Following Wicherts and Dolan (2010), any firm conclusion regarding the interpretation of gender differences in intelligence testing must pass through a test of measurement invariance that takes into accounts the consequences of non-invariant intercepts on the latent mean differences. Accordingly, we aimed to address the extent to which gender differences in subtests scores and indices could be attributed to genuine differences in the ability factors, or gender-specific processes unrelated to cognitive abilities. Because the WAIS-III has never been translated into Italian, different from previous research, we analyzed the WAIS-IV that is the latest and currently used edition.

The literature suggests that the gender gap in cognitive abilities has been narrowing across generations (e.g., Hyde, 2016). The best comparable estimate of the cognitive abilities of past generations that we have in Italy comes from the normative sample of the WAIS-R. While the Italian WAIS-IV was published in 2013, the WAIS-R was released in 1997. Therefore, at least 16 years are elapsed between the two normative samples. Our analyses are primarily focused on the WAIS-IV, using the WAIS-R data only for comparison. This comparison could help quantify the gender bias in specific subtests common to both editions that the literature discussed in the context of gender differences, such as Information and Arithmetic (e.g., Arceneaux, Cheramie, & Smith, 1996; Held, Alderton, Foley, & Segall, 1993; Irwing, 2012; Lynn & Dai, 1993; Lynn, Fergusson, & Horwood, 2005; Snow & Weinstock, 1990).

2. Method

2.1. Participants

The present study used data from the Italian normative samples of two editions of the Wechsler Adult Intelligence Scale (WAIS-IV and WAIS R). All participants were from the general population, free of neurological, psychiatric, or sensory disabilities, and were recruited from local and national companies, associations, senior community centers, academic institutions, and by word of mouth. The Wechsler scales were administered in a quiet room at home or participants' community centers in individual sessions lasting between 60 and 90 min. The WAIS-IV sample (Orsini & Pezzuti, 2013, 2015) included 2174 healthy adults (1072 men and 1102 women) selected as a quota sample of the Italian population regarding gender, geographical area, and socio-cultural level. Participants' age ranged from 16 to 90 years (M = 50.9 years, SD = 24.4 years). Men and women subsamples did not differ regarding age (eta-squared = 0.00) and years of education (eta-squared = 0.00) (see Table 1 for the joint distribution of Gender, Age, and Education). The WAIS-R sample (Orsini & Laicardi, 1997, 2003) included 2798 healthy adults (1399 women and 1399 men), ranging from 16 to 85 (M = 49.9 years, SD = 23.0 years). Men and women subsamples did not differ regarding age (eta-squared = 0.00) and years of education (eta-squared = 0.00). As for the WAIS-IV, Table 1 reported the joint distribution of Gender, Age, and Education for the WAIS-R. Detailed characteristics of both samples are available in the test manuals.

Table 1
Joint distribution of Gender, Age and Education for WAIS-R and WAIS-IV standardization samples.

Age range	Education	WAIS-IV			WAIS-R		
		Women	Men	Total	Women	Men	Total
16–19	Middle School	110	109	219	117	109	226
	High School	26	31	57	15	12	27
	Total	136	140	276	132	121	253
20–29	Middle School	9	14	23	34	40	74
	High School	103	115	218	244	243	487
	University	73	47	120	40	46	86
	Total	185	176	361	318	329	647
30–44	Eleme ntary	1	2	3	12	9	21
	Middle School	22	18	40	43	36	79
	High School	66	85	151	73	77	150
	University	74	60	134	37	43	80
	Total	163	165	328	165	165	330
	Elementary	3	2	5	27	28	55
45–54	Middle School	22	16	38	15	25	40
	High School	49	41	90	40	29	69
	University	25	15	40	18	18	36
	Total	99	74	173	100	100	200
	Elementary	7	1	8	33	30	63
55–64	Middle School	18	27	45	13	28	41
	High School	44	30	74	48	23	71
	University	18	23	41	6	19	25
	Total	87	81	168	100	100	200
	Elementary	302	249	551	435	360	795
> 65	Middle School	60	77	137	88	154	242
	High School	53	69	122	54	52	106
	University	17	41	58	7	18	25
	Total	432	436	868	584	584	1168
Total	Elementary	313	254	567	508	427	935
	Middle School	241	261	502	309	392	701
	High School	341	371	712	474	436	910
	University	207	186	393	108	144	252
	Total	1102	1072	2174	1399	1399	2798

2.2. Instruments

The WAIS-IV (Orsini & Pezzuti, 2013, 2015; Wechsler, 2008) is a clinical tool, administered individually, to assess the cognitive abilities of individuals aged from 16 years and 0 months to 90 years. It consists of the following 15 subtests (10 core and five supplementary): Similarities, Vocabulary, Information, Comprehension, Block Design, Matrix Reasoning, Visual Puzzles, Figure Weights, Figure Completion, Digit Span, Arithmetic, Letter-Number Sequencing, Symbol Search, Coding, and Cancellation. In the standardization sample, the 15 subtests were administered following the order proposed in the original manual. Letter-Number Sequencing, Figure Weights, and Cancellation are supplemental WAIS-IV subtests, originally administered only up to 69 years of age. Because there were no empirically supported reasons for not administering these subtests to the elderly (Wechsler, 2008), and because the factorial invariance of the 15 WAIS-IV subtests between Italian adults and elderly was recently demonstrated (see Pezzuti & Rossetti, 2017a, 2017b), we used all the subtests up to 90 years old. The WAIS-IV yields four composite indices, each corresponding to a specific area of cognitive functioning (i.e., Verbal Comprehension Index, or VCI; Perceptual Reasoning Index, or PRI; Working Memory Index, or WMI; and Processing Speed Index, or PSI). The weighted sum of the four indices provides a Full-Scale IQ (FSIQ) reflecting the overall cognitive functioning. The WAIS-R (Orsini & Laicardi, 1997, 2003; Wechsler, 1981) was comprised of 11 subtests (i.e., Information, Picture Completion, Digit Span, Vocabulary, Block Design, Arithmetic, Comprehension, Digit Symbol, Similarities, Picture Arrangement, Object Assembly) that were administered in the order proposed in the original manual. The WAIS-R yields a Verbal IQ (VIQ), a Performance IQ (PIQ), and a Full-Scale IQ (FSIQ).

2.3. Data analysis

Although we used the Italian normative data, gender sub-samples could not be representative of the male and female populations (e.g., Hunt & Madhyastha, 2008). For instance, there might be gender by cohort interactions on educational attainment, with males more highly educated in the older generations and females averaging more years of education in the most recent decade (Parro, 2012). To quantify potential biases in the recruitment process, we assessed how the joint distribution of Gender, Age, and Education in the two normative samples (reported in Table 1) deviated from the expected frequencies based on the corresponding distributions Italian census data. Accordingly, we developed sample weights to adjust for under- or over-representations of particular cells. The weight variables for WAIS-R and WAIS-IV data were based on the product of the marginal proportions for gender, age, and education in the 1991 and 2011 divided by the observed marginal proportion in the two normative samples (collected in 1997 and 2013, respectively). All the analyses used age-standardized subtest scores and were carried out twice, using unweighted and census-weighted data. While we disclose both types of results in tables and figures, we only commented when they differed substantially.

Before examining the measurement invariance of the WAIS-IV by gender (and of the WAIS-R, for comparison), we assessed gender differences in subtest scores, indices, and FSIQs. We classified the effect sizes according to Hyde's (2005) guidelines for interpreting gender differences: close-to-zero (Cohen's $d < 0.10$), small ($0.11 < d < 0.35$), moderate ($0.36 < d < 0.65$), and large ($0.66 < d < 1.00$). Achieving measurement invariance for a second-order model requires a series of successive steps in which a baseline model is subsequently modified by adding restrictions to specific sets of parameters (see Rudnev, Lytkina, Davidov, Schmidt, & Zick, 2018). Consistent with the WAIS-IV scoring system, the baseline model tested was a second-order one with a general ability factor on the top and four first-order factors, which reflected Verbal Comprehension, Perceptual Reasoning, Working Memory, and Processing Speed abilities, respectively. The latent factors were measured using core and supplemental subtests to clarify the substantive meaning of abilities assessed by WAIS-IV (e.g., Benson, Hulac, & Kranzler, 2010). Consistent with the WAIS-R scoring method, we used a baseline model with one general factor and two first-order factors, reflecting Verbal and Performance abilities.

MGCFA-s were conducted using the "lavaan" (Rosseel, 2012) and "lavaan.survey" (Oberski, 2014) packages for R with Maximum Likelihood Estimators. Because of the relatively large sample size, the model chi-square ($\chi 2$) is likely to turn out statistically significant even when the misfit of the model is negligible (Chen, 2007). We assessed model fit using the Comparative Fit Index (CFI), Tucker-Lewis index (TLI), Root-Mean-Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). The accepted standards for CFI and TLI are values higher than 0.90 and 0.95 for an acceptable and a good fit, respectively. For RMSEA and SRMR, values lesser than 0.06 and 0.08 are widely accepted (e.g., Schermelleh-Engel, Moosbrugger, & Müller, 2003).

We started the MGCFA testing a configuration invariance analysis aimed at establishing the fit of the baseline model. In this model, the factor loadings of the first-order and second-order factors were free, but one per factor was fixed to 1; all the subtests intercepts were free, but one per factor was fixed to 0; the latent means of the first-order factors were free, and the latent mean of the second-order factor that was fixed to 0. To test the metric invariance of the first-order factors, we constrained the factor loadings of the first-order factors to be equal between groups. The following model added equality constraints to the factor loadings of the second-order factors. The residual terms were set equal in subsequent analysis. Once the metric and residual terms invariance were achieved, we tested the first-order scalar invariance, imposing equality constraints on the subtest intercepts, while the latent

Table 2
Gender differences in WAIS-IV core and supplemental subtests, and indices using unweighted and census-weighted standardization data.

Subtests	Unweighted data					Weighted data				
	t (2173)	*p*	Cohen's d	95% CI for d		t (2173)	*p*	Cohen's d	95% CI for d	
Block Design	−6.25	< .001	−0.27	−0.35	−0.18	−6.59	< .001	−0.29	−0.36	−0.19
Similarities	0.85	.401	0.04	−0.05	0.12	0.67	.505	0.03	−0.06	0.11
Digit Span	−4.49	< .001	−0.19	−0.28	0.11	−4.28	< .001	−0.18	−0.27	−0.10
Matrix Reasoning	−3.79	< .001	−0.16	−0.25	−0.08	−4.45	< .001	−0.19	−0.28	−0.11
Vocabulary	1.65	.101	0.07	−0.01	0.15	0.97	.330	0.04	−0.04	0.13
Arithmetic	−11.51	< .001	−0.49	−0.58	0.41	−10.88	< .001	−0.47	−0.55	−0.47
Symbol Search	−1.17	.243	−0.05	−0.13	0.03	−2.02	.044	−0.09	−0.17	−0.00
Visual Puzzles	−6.19	< .001	−0.27	−0.35	−0.18	−6.89	< .001	−0.30	−0.38	−0.21
Information	−7.09	< .001	−0.30	−0.39	−0.22	−6.82	< .001	−0.29	−0.38	−0.21
Coding	1.51	.134	0.06	−0.02	0.15	0.57	.571	0.02	−0.06	0.11
Letter-Number Sequencing	−4.66	< .001	−0.20	−0.28	−0.12	−5.04	< .001	−0.22	−0.30	−0.13
Figure Weights	−7.61	< .001	−0.33	−0.41	0.24	−7.52	< .001	−0.32	−0.41	−0.24
Comprehension	−2.70	.01	−0.12	−0.20	−0.03	−3.27	.001	−0.14	−0.23	−0.06
Cancellation	0.15	.885	0.01	−0.08	0.09	0.15	.880	0.01	−0.08	0.09
Picture Completion	−3.49	< .001	−0.15	−0.23	−0.07	−3.28	.001	−0.14	−0.23	−0.06
Indices										
Verbal Comprehension	−1.70	.096	−0.07	−0.16	0.01	−1.95	.052	−0.08	−0.17	0.00
Perceptual Reasoning	−6.63	< .001	−0.28	−0.37	−0.20	−7.29	< .001	−0.31	−0.40	−0.23
Working Memory	−9.00	< .001	−0.39	−0.47	−0.30	−8.55	< .001	−0.37	−0.45	−0.28
Processing Speed	0.16	.870	0.00	−0.08	0.09	0.82	.412	−0.04	−0.05	0.12
Full Scale IQ	−5.15	< .001	−0.22	−0.31	−0.14	−5.55	< .001	−0.24	−0.32	−0.15

Note: Supplemental subtests are in italics; According to Hyde's guidelines for interpretation of gender differences (2005), d values equal to or less than 0.10, between 0.11 and 0.35, and between 0.36 and 0.65 represent null, small, and moderate effect sizes, respectively.

means of the first-order factors were freely estimated. Full scalar invariance is challenging to achieve, especially in the light of substantial gender differences in cognitive abilities that were found in previous studies of the WAIS-IV (e.g., Daseking et al., 2017). Similar principles apply to the invariance latent means of the first-order factors and the latent mean difference of the second-order factor.

A chi-square difference test can formally test each level of measurement invariance. However, because this test is overly sensitive to sample size, other additional measures of change in the goodness of fit have been suggested. Accordingly, we used ΔCFI and ΔNCI (i.e., the difference in MacDonald Non-centrality Index), which are deemed the most sensitive indices to detect lack of invariance. In particular, a ΔCFI not larger than 0.002 supports substantial equivalence in relative model fit. Cutoff values for ΔNCI depend on the number of factors and the number of subtests, varying from 0.008 for 15 subtests and four factors (as in the analysis of WAIS-IV data) to 0.007 for 11 subtests and two factors (as in the analysis of WAIS-R data). Because - to our knowledge - no exact cut-off values have been provided for invariance analysis of second-order factor models, we set to the most conservative value of 0.007 for both 15 and 11 subtests (see table 12 in Meade, Johnson, & Braddy, 2008). This value corresponds to the 95th percentile of ΔNCI in recent empirical simulation studies of lack of measurement invariance (Kang, McNeish, & Hancock, 2016). If the measurement invariance was not supported at one of the different levels (metric, residual, or scalar), the model could be modified based on the modification indices (MI). These indices helped to identify the cause of the misfit and whether the partial invariance could be achieved.

3. Results

For both unweighted and census-weighted data, men performed significantly better than women in the core subtests of the WAIS-IV, like Block Design, Digit Span, Matrix Reasoning, Arithmetic, Visual Puzzles, and Information. There was a significant male advantage in supplemental subtests, like Letter-Number Sequencing, Figure Weights, Comprehension, and Picture Completion. According to Hyde guidelines for interpreting gender differences (2005), only the effect size for the Arithmetic subtest has to be considered a moderate difference. Otherwise, the effect sizes were small for the remaining statistically significant subtests. Regarding indices, we found gender differences in PRI, WMI, and FSIQ with a moderate effect size for the working memory score and small ones for the remaining comparisons (see Table 2).

Table 3 reported corresponding analyses using the WAIS-R. Unweighted comparisons revealed that men outperformed women in all subtests but Vocabulary and Coding. The Similarities subtest was not statistically significant by gender using census-weighted data. According to Hyde (2005), gender differences in Arithmetic, Information, and Block Design exceeded the threshold for moderate effect size. The remaining statistically significant differences were small. Men also obtained higher IQ scores than women with a small-medium effect size.

The previous analyses suggested that gender differences existed in specific subtests and cognitive abilities when we analyzed the WAIS-IV standardization sample, while they appeared more pronounced and pervasive using WAIS-R data. Solid dark bars in Fig. 1 display the absolute value of the effect sizes corresponding to the mean difference between men and women for each subtest in each edition of the WAIS. The diagonal stripes bars show the factor loadings, estimated in the male group without equality constraints. The overall fit of these models corresponds to that of the baseline models in MGCFA (i.e., M0 in Tables 5 and 6). After identifying the subtest with the highest loading for each factor, in a subsequent analysis we set the measurement unit of each factor as equal to the observed effect size for that subtest (starred in Fig. 1). This procedure allows us to compare the effect-size of each subtest against its factor using a common metric. It supports the interpretation of intercept differences that might result from MGCFA (for details see Wicherts & Dolan, 2010). Indeed, when the factor loading and the effect size are about the same (i.e., collinear), the observed effect size at the subtest level reflects a genuine gender difference at the construct level. Conversely, a striking discrepancy between factor loadings and effect sizes in Fig. 1, indicates a measurement bias due to construct-irrelevant variance components.

Inspecting both unweighted and census-weighted analyses carried out on the WAIS-IV (Fig. 1a and 1b, respectively), the Information and Comprehension subtests exhibited an effect size larger than expected to result from a genuine gender difference in the mean of the Verbal Comprehension factor. Their loadings were about as large as those of the other verbal subtests, but their effect sizes were much larger.

Table 3
Gender differences in WAIS-R subtests and indices using unweighted and census-weighted standardization data.

Subtests	Unweighted data					Weighted data				
	t (2796)	*p*	Cohen's d	95% CI for d		t (2796)	*p*	Cohen's d	95% CI for d	
Information	−12.36	< .001	−0.47	−0.54	−0.39	−10.40	< .001	−0.39	−0.47	−0.32
Digit Span	−6.88	< .001	−0.26	−0.33	−0.19	−7.35	< .001	−0.28	−0.35	−0.20
Vocabulary	−2.10	.043	−0.08	−0.15	−0.00	−1.49	.140	−0.06	−0.13	0.02
Arithmetic	−16.15	< .001	−0.61	−0.69	−0.53	−15.02	< .001	−0.57	−0.64	−0.49
Comprehension	−6.07	< .001	−0.23	−0.30	−0.16	−5.59	< .001	−0.21	−0.29	−0.14
Similarities	−1.99	.050	−0.08	−0.15	−0.00	0.39	.700	0.01	−0.06	0.09
Picture Completion	−8.29	< .001	−0.31	−0.39	−0.24	−8.25	< .001	−0.31	−0.39	−0.24
Picture Arrangement	−5.91	< .001	−0.22	−0.30	−0.15	−5.73	< .001	−0.22	−0.29	−0.14
Block Design	−10.99	< .001	−0.42	−0.49	−0.34	−10.48	< .001	−0.40	−0.47	−0.32
Object Assembly	−6.40	< .001	−0.24	−0.32	−0.17	−4.95	< .001	−0.19	−0.26	−0.11
Coding	−1.41	.162	−0.05	−0.13	0.02	0.85	.390	0.03	−0.04	0.11
IQ scores										
Verbal IQ	−9.45	< .001	−0.36	−0.43	-0.28	-8.06	< .001	-0.30	-0.38	0.23
Performance IQ	-7.49	< .001	-0.28	-0.36	-0.21	-6.66	< .001	-0.25	-0.33	-0.18
Full Scale IQ	-9.49	< .001	-0.36	-0.43	-0.28	-8.37	< .001	-0.32	-0.39	0.24
Factor-derived indices										
Verbal	-6.63	< .001	-0.25	-0.33	-0.33	-5.05	< .001	-0.22	-0.30	-0.13
Perceptual Organization	-10.29	< .001	-0.39	-0.46	-0.46	-9.60	< .001	-0.41	-0.49	-0.33
Freedom from distractibility	-13.46	< .001	-0.51	-0.58	-0.58	-13.13	< .001	-0.57	-0.65	0.48

Note: Supplemental subtests are in italics; According to Hyde's guidelines for interpretation of gender differences (2005), d values equal to or less than 0.10, between 0.11 and 0.35, and between 0.36 and 0.65 represent null, small, and moderate effect sizes, respectively.

Similarly, the Arithmetic subtest loaded on the Working Memory factor about to the same extent as Digit Span and Letter-Number Sequencing (Fig. 1a and 1b), but its effect size was almost twice as much as the other subtests. For the Processing Speed factor, the loadings were about the same for all subtests, while the corresponding effect sizes were negligible (especially for coding and cancellation subtests in the analysis of census weighted data). For the Perceptual Reasoning factor, the effect sizes never exceeded the scaled factor loading value. As it regards the WAIS-R (Fig. 1c and 1d, for unweighted and weighted data, respectively), all the verbal subtests but Similarities and Vocabulary

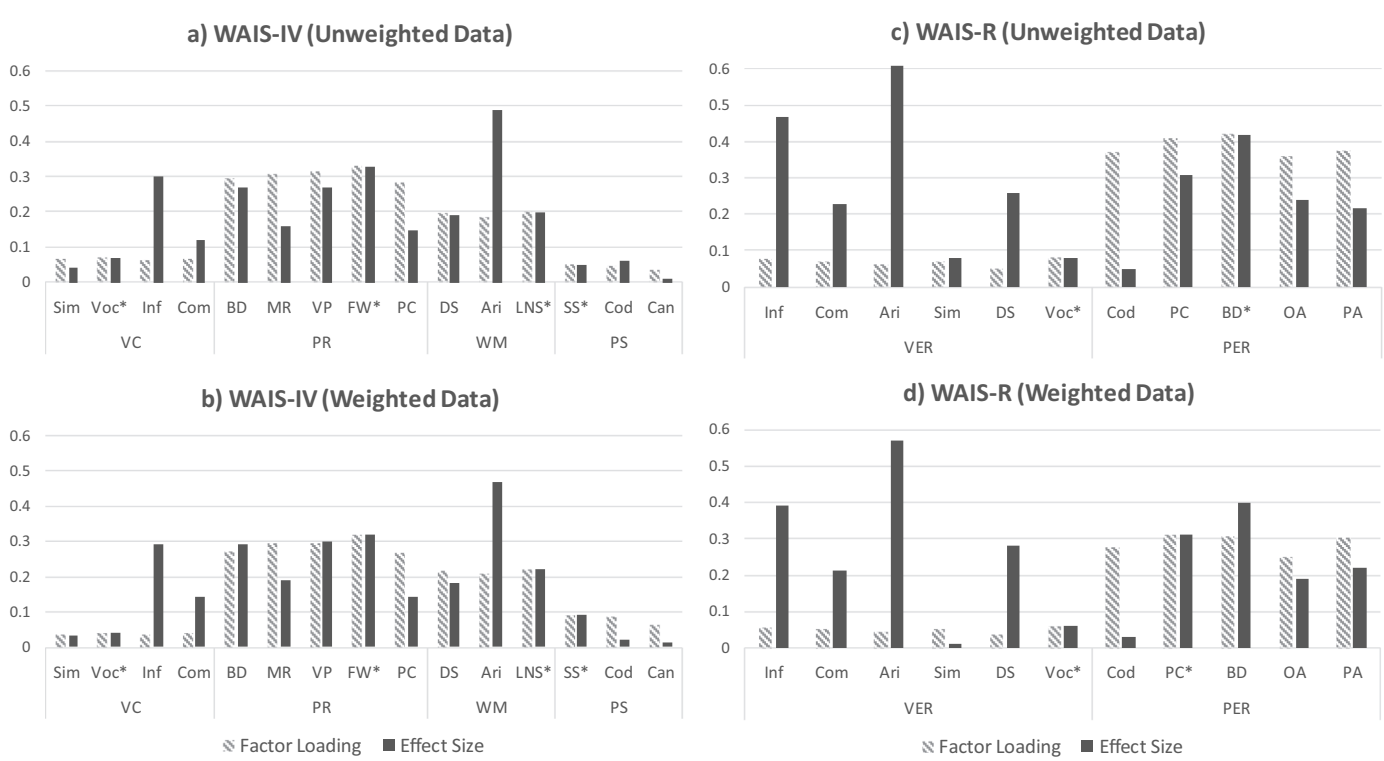

Figure 1. Factor loadings and effect sizes per subtest using WAIS-IV and WAIS-R unweighted and census-weighted data.
Legend: Ari = Arithmetic; BD = Block Design; Can = Cancellation; Cod = Coding; Com = Comprehension; DS = Digit Span; FW = Figure Weights; Inf = Information; LNS = Letter-Number Sequencing; MR = Matrix Reasoning; OA = Object Assembly; PA = Picture Arrangement; PC = Picture Completion; Sim = Similarities; SS = Symbol Search; VP = Visual Puzzles; Voc = Vocabulary; VC = Verbal Comprehension; PR = Perceptual Reasoning; WM = Working Memory; PS = Processing Speed; VER = Verbal; PER = Performance. * = indicates the subtest used to give a metric to the corresponding factor. Solid dark bars display the absolute value of the effect sizes corresponding to gender differences in each subtest. The diagonal stripes bars show the unstandardized factor loadings, estimated in the male group without equality constraints.

Table 4
Tests of measurement invariance by gender for WAIS-IV using unweighted and census-weighted standardization data

Models and constraints	Analysis	χ2(df)	Δχ2(df)	CFI	ΔCFI	NCI	ΔNCI	TLI	RMSEA	90%CI	SRMR
M0. Configural invariance	Unweighted	836.21(172)		.959		.858		.950	.060	.056-.064	.039
	Weighted	878.29(172)		.958		.850		.949	.061	.057-.066	.039
M1. Metric invariance FOF	Unweighted	850.09(183)	13.88(11)***	.959	.000	.858	.000	.953	.058	.054-.062	.041
	Weighted	896.33(183)	13.99(11)***	.958	.000	.849	.001	.951	.060	.056-.064	.041
M2. Metric invariance SOF	Unweighted	851.35(186)	1.27(3)***	.959	.000	.858	.000	.954	.057	.053-.061	.041
	Weighted	898.51(186)	1.72(3)***	.958	.000	.849	.000	.952	.059	.056-.063	.042
M3. Residual invariance	Unweighted	886.65(201)	35.30(15)***	.958	.001	.854	.004	.956	.056	.052-.060	.042
	Weighted	936.88(201)	24.85(15)***	.956	.002	.844	.005	.954	.058	.054-.062	.043
M4. Scalar invariance FOF	Unweighted	1124.45(212)	237.80(11)***	.944	.014	.811	.043	.944	.063	.059-.067	.045
	Weighted	1146.18(212)	220.16(11)***	.945	.011	.807	.038	.945	.064	.070-.067	.046
M4a. Scalar invariance FOF[1]	Unweighted	1026.06(211)	139.41(10)***	.950	.008	.829	.025	.950	.060	.056-.063	.044
	Weighted	1062.29(211)	118.02(10)***	.950	.007	.822	.022	.950	.061	.057-.065	.045
M4b. Scalar invariance FOF[1,2]	Unweighted	948.06(210)	61.41(9)***	.954	.004	.844	.010	.954	.057	.053-.061	.043
	Weighted	999.91(210)	54.44(9)***	.953	.005	.834	.010	.953	.059	.055-.063	.044
M4c. Scalar invariance FOF[1,2,3]	Unweighted	915.64(209)	28.99(8)***	.956	.002	.850	.004	.956	.056	.052-.059	.043
	Weighted	965.69(209)	23.66(8)***	.955	.001	.840	.004	.955	.058	.054-.061	.044

Legend: df = degrees of freedom; Δχ2 = chi square difference test; CFI = Comparative Fit Index; ΔCFI = difference in CFI from the logically antecedent model; NCI = McDonald Non-Centrality Index; ΔNCI = difference in NCI from the logically antecedent model; TLI = Tucker-Lewin Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual; FOF = First-order factors; SOF = Second-order factors.
Note: $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.
[1] Information intercept freed.
[2] Arithmetic intercept freed.
[3] Comprehension intercept freed.

exhibited a effect size considerably larger their factor loading. For the Performance factor, the effect size of the Block Design subtest exceeded the scaled factor loading value, but only in the weighted data. The differences observed in Fig. 1 for Information, Comprehension, and Arithmetic suggested that there might be corresponding differences in the intercepts of these subtests that not only might undermine the measurement invariance of verbal and working memory abilities but also might indicate measurement bias. Before addressing this issue, we have formally examined the measurement invariance of the WAIS-IV, comparing it to a parallel analysis on the WAIS-R.

Table 4 reports the fit indices and hypothesis tests regarding the measurement invariance of the WAIS-IV. The configural invariance model (M0) was an excellent fit for all indices, and both unweighted and census-weighted data. The model testing the metric invariance of the first-order factors (M1) was also an excellent fit. The χ2 difference test showed no significant decline in model fit relative to M0. Extending the test of metric invariance to second-order factors (M2) also yielded an excellent fit. Indeed, the χ2 difference test between M1 and M2 was not statistically significant. Before examining the scalar invariance hypothesis, we also tested the equivalence of residual terms (M3). Although the χ2 difference test between M2 and M3 was statistically significant in the analysis of unweighted data, the ΔCFI and ΔNCI were negligible in both the analyses (Kang et al., 2016).

The scalar invariance model (M4) was a significant worsening in model fit compared to M3. Therefore, the intercept equality assumption was not supported. Using modification indices, we conducted a specification search aimed to detect the intercepts that had to be let free to vary to meet the partial measurement invariance. Accordingly, the intercept of the Arithmetic subtest was relaxed. Nevertheless, the modified model (M4a) was a significant loss of fit compared to M3. Next, we released the constraint imposed on the intercept of the Information subtest. The modified model (M4b) still was a significant loss of fit compared to M3. Therefore, a subsequently modified model (M4c) relaxed the equality constraint for the intercept of the Comprehension subtest. Although the Δχ2 between M4c and M3 was significant, the ΔCFI and the ΔNCI were acceptable. This finding prompted us to end the specification search, supporting the equality of model intercepts for all subtests but Arithmetic, Information, and Comprehension (see Fig. 2).

Because the modified model M4c also supported the scalar invariance of the Perceptual Reasoning and Processing Speed factors, we could meaningfully examine the difference in latent means for these factors. Descriptively, the unweighted unstandardized latent means for the Perceptual Reasoning factor were 9.59 (SD = 0.73) and 10.36 (SD = 1.01) for women and men, respectively (M = 9.87; SD = 0.70 and M = 10.70, SD = 1.08, using census-weighted data). For the Processing Speed factor, using unweighted data, the latent means were 9.98 (SD = 1.67) and 9.97 (SD = 1.67) for women and men, respectively (M = 10.08; SD = 1.70 and M = 10.18, SD = 1.71, using census-weighted data). To test the statistical significance of gender differences, we first imposed equality constraints to the latent mean of the Processing Speed factor, which appeared to yield similar estimates for men and women. The chi-square difference test (with 1 df) relative to M4c was not significant both in the analysis of unweighted and census-weighted data ($\Delta\chi2$-s = 0.01 and 0.53; p-s = .921 and .465, respectively). Next, we imposed equality to the latent mean of the Perceptual Reasoning factor. The chi-square difference test (with 1 df) relative to M4c was significant in both the analyses of unweighted and census-weighted data ($\Delta\chi2$-s = 52.84 and 76.29, respectively; both p-s < .001).

Although there is a divide on whether partial scalar invariance could be enough for meaningful cross-group comparisons (e.g., De Beuckelaer & Swinnen, 2018; Kuha & Moustaki, 2015), we also investigated the latent means of the Working Memory and Verbal Comprehension factors to aid the interpretation of the data. For the Working Memory factor, using unweighted data, the latent means were 9.70 (SD = 0.96) and 10.28 (SD = 1.04) for women and men, respectively (M = 9.93; SD = 0.93 and M = 10.53, SD = 0.96, using census-weighted data). Last, the estimates for the Verbal Comprehension factor were the following: women's M = 10.09 (SD = 1.53) and men's M = 9.91 (SD = 1.58) using unweighted data; women's M = 10.44 (SD = 1.49) and men's M = 10.32 (SD = 1.55) using census-weighted data. Equalizing the latent means of the Working Memory factor, the analysis yielded a poorer fit than M4c ($\Delta\chi2$-s = 25.91 and 25.98, for unweighted and census-weighted data, respectively; both p-s < .001). By contrast, equalizing the latent means of the Verbal Comprehension factor, the analysis yielded the same fit as M4c ($\Delta\chi2$-s = 2.14 and 0.69, p-s = .144 and .405, respectively for unweighted and census-weighted data).

Considering the findings reported so far for the WAIS-IV, a logical

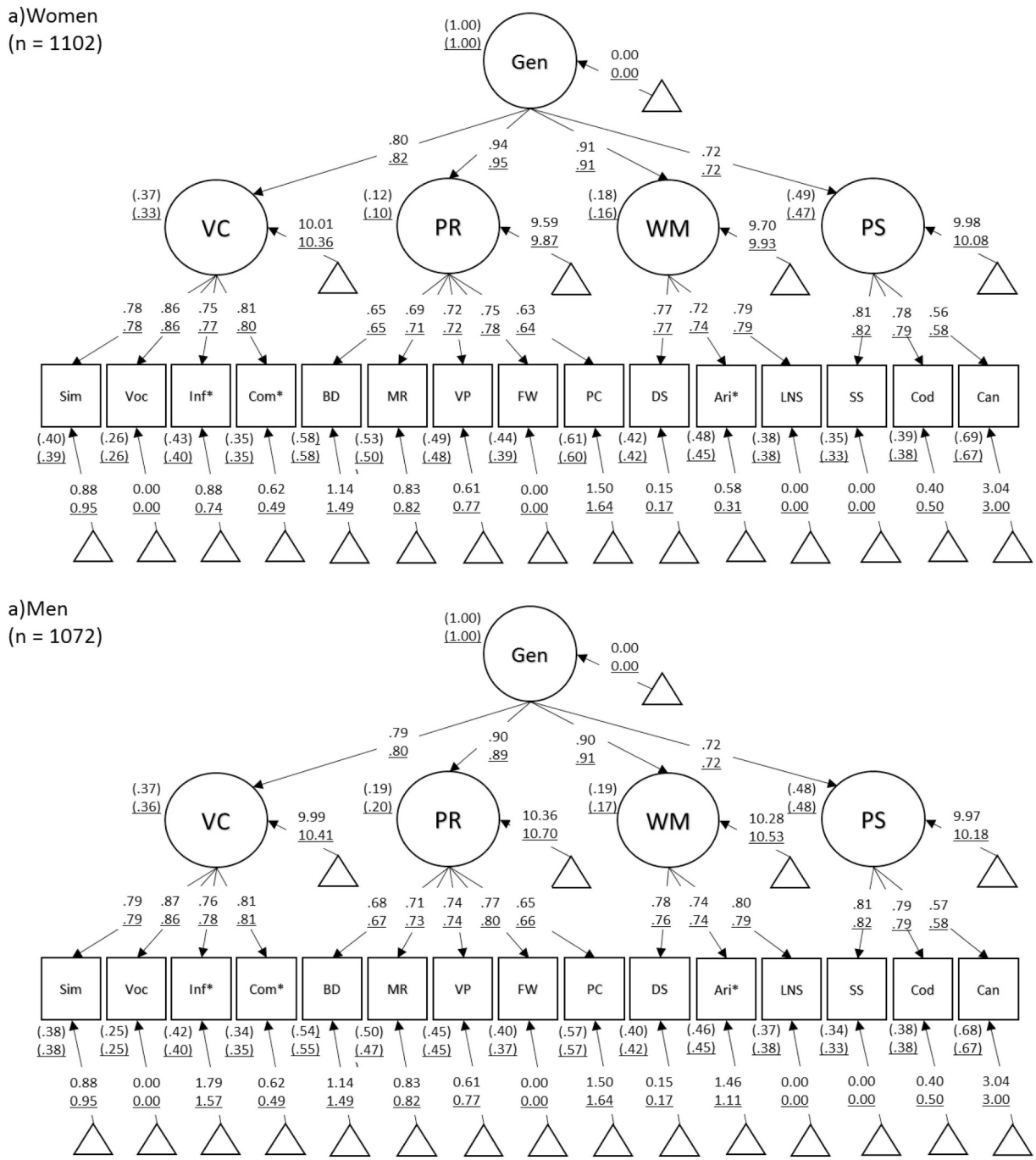

Figure 2. Path diagram of partial measurement invariance model (M4c) of the WAIS-IV using unweighted and census-weighted standardization data.
Legend: Ari = Arithmetic; BD = Block Design; Can = Cancellation; Cod = Coding; Com = Comprehension; DS = Digit Span; FW = Figure Weights; Inf = Information; LNS = Letter-Number Sequencing; MR = Matrix Reasoning; PC = Picture Completion; Sim = Similarities; SS = Symbol Search; VP = Visual Puzzles; Voc = Vocabulary; VC = Verbal Comprehension; PR = Perceptual Reasoning; WM = Working Memory; PS = Processing Speed; Gen = General factor.
Note: The intercept of starred subtests was freed between gender groups. Factor loadings are standardized. Underlined coefficients refer to census-weighted data.

question was whether gender differences in Arithmetic, Information, and Comprehension subtests were of some practical importance for the assessment of Verbal Comprehension and Working Memory abilities. According to Wicherts and Dolan (2010), if there are enough invariant subtests for an ability factor, one can estimate the expected mean difference of a specific subtest based on the product of the latent factor mean difference times the factor loading of that subtest. By comparing this expected mean difference - reflecting the difference in the latent ability only - to the actual mean difference in the subtest score, one can obtain an estimate of the bias at the subtest level. Indeed, the term bias is used to refer to any group difference that cannot be accounted for by corresponding differences on the construct that the test aims to measure. Following this procedure, we estimated a bias in favor of men of 1.05 and 0.51 for Information and Comprehension, respectively (1.03 and 0.59 using census-weighted data), and of 0.89 (0.85 using census-weighted data) for Arithmetic. Because the age-standardized subtest scores have a standard deviation of 3.00, the bias we observed in the Information subtest in favor of men would correspond to a z-score of +0.34, which is an overestimate of +13% in terms of percentile rank. For the Arithmetic and Comprehension subtests, the bias would

Table 5
Tests of measurement invariance by gender for WAIS-R using unweighted and census-weighted standardization data

Models and Constraints	Analysis	χ2(df)	Δχ2(df)	CFI	ΔCFI	NCI	ΔNCI	TLI	RMSEA	90%CI	SRMR
M0. Configural invariance	Unweighted	944.92 (86)		.931		.857		.912	.084	.080-.089	.045
	Weighted	976.55 (86)		.928		.853		.909	.086	.081-.091	.047
M1. Metric invariance FOF	Unweighted	964.93 (95)	20.01(9)***	.930	.001	.856	.002	.919	.081	.076-.086	.048
	Weighted	1008.80 (95)	19.11(9)***	.927	.001	.849	.003	.915	.083	.078-.088	.052
M2. Metric invariance SOF	Unweighted	965.39 (96)	0.46(1)***	.930	.000	.856	.000	.920	.080	.076-.085	.048
	Weighted	1009.28 (96)	0.31(1)***	.927	.000	.849	.000	.916	.082	.078-.087	.052
M3. Residual invariance	Unweighted	992.76 (107)	27.37(11)***	.929	.001	.854	.003	.927	.077	.073-.081	.049
	Weighted	1022.50 (107)	7.06(11)***	.926	.001	.849	.000	.924	.078	.074-.083	.053
M4. Scalar invariance FOF	Unweighted	1443.86 (116)	451.09(9)***	.894	.035	.789	.065	.899	.090	.086-.095	.060
	Weighted	1508.61(116)	478.05(9)***	.888	.038	.780	.069	.894	.093	.088-.097	.064
M4a. Scalar invariance FOF[1]	Unweighted	1263.47(115)	270.70(8)***	.908	.021	.814	.039	.912	.084	.080-.089	.055
	Weighted	1332.14(115)	259.73(8)***	.902	.024	.805	.045	.906	.087	.083-.091	.060
M4b. Scalar invariance FOF[1,2]	Unweighted	1121.73(114)	128.97(7)***	.919	.010	.835	.018	.922	.079	.075-.084	.053
	Weighted	1211.99(114)	141.79(7)***	.912	.015	.821	.028	.915	.083	.079-.087	.059
M4c. Scalar invariance FOF[1,2,3]	Unweighted	1067.47(113)	74.71(6)***	.924	.006	.843	.010	.926	.078	.073-.082	.051
	Weighted	1130.36(113)	73.64(6)***	.918	.008	.834	.015	.920	.080	.076-.085	.056
M4d. Scalar invariance FOF[1,2,3,4]	Unweighted	1038.31(112)	45.54(5)***	.926	.003	.847	.006	.927	.077	.073-.081	.050
	Weighted	1084.93(112)	41.26(5)***	.922	.005	.840	.009	.923	.079	.075-.083	.054
M4e. Scalar invariance FOF[1,2,3,4,5]	Unweighted	1010.64(111)	17.87(4)***	.928	.001	.851	.002	.929	.076	.072-.080	.050
	Weighted	1048.75(111)	16.51(4)***	.925	.002	.846	.003	.925	.078	.073-.082	.054

df = degrees of freedom; Δχ2 = chi square difference test; CFI = Comparative Fit Index; ΔCFI = difference in CFI from the logically antecedent model; NCI = McDonald Non-Centrality Index; ΔNCI = difference in NCI from the logically antecedent model; TLI = Tucker-Lewin Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual; FOF = First-order factors; SOF = Second-order factors.
Note: $*p < .05$; $**p < .01$; $***p < .001$.
[1] Arithmetic intercept freed.
[2] Information intercept freed.
[3] Coding intercept freed.
[4] Digit Span intercept freed.
[5] Comprehension intercept freed.

correspond to an overestimation of +11% and +8% in male performance, respectively.

For comparison, we also tested the measurement invariance of the WAIS-R. The configural, metric, and residual terms invariance hypotheses were supported (see M0-M3 in Table 5). The scalar invariance of the first-order factors (M4) was instead rejected. As for the WAIS-IV, we conducted a specification search based on modification indices. Unlike WAIS-IV, we had to relax the equality constraints imposed on the intercepts of five subtests in subsequent tests of model fit before achieving the partial scalar invariance (see M4a–M4e in Table 5). Indeed, Arithmetic, Information, Coding, Digit Span, and Comprehension caused severe misfit to the full scalar invariance model. Although M4e supported the partial scalar invariance of the WAIS-R, the intercepts of most subtests had to be released. This result precluded any unbiased test of the latent mean differences.

Again, we assessed the practical importance of intercept differences. We estimated a bias in favor of men of 1.12, 0.48, 1.58, and 0.61 (in age-standardized subtest units), respectively for Information, Comprehension, Arithmetic, and Digit Span (1.02, 0.54, 1.52, and 0.74 using census-weighted data), and of 0.73 (0.93 using census-weighted data) for Coding. In percentile ranks units, we estimated the bias to lead to an overestimation of male performance around +13% in Information, +7% in Comprehension, +19% in Arithmetic, and +10% in Digit Span. For Coding, women's performance was overestimated by +12%.

The previous analyses have shown that neither the WAIS-IV nor the WAIS-R achieved the full scalar invariance of the first-order factors. Gender differences seemed to affect the most the verbal scales of the two editions (VCI and VIQ, respectively). It is, however, well-established that Arithmetic and Digit Span, are no longer considered verbal subtests in the WAIS-IV. Considering that some subtests are edition-specific (e.g., Figure Weights in the WAIS-IV and Object Assembly in WAIS-R), we carried out a new analysis using only the common subtests. We specified a hierarchical model with a second-order general factor and three first-order factors, representing Verbal, Perceptual, and Working Memory abilities. As one can see from Table 6, no substantial differences emerged between unweighted and census-weighted data, except for the fit-change of M4d, WAIS-R, census-weighted data. The metric and residual terms invariances (M1-M3) were supported for both scales. The scalar invariance of the first-order factors was rejected (M4). The intercepts of Information and Comprehension subtests had to be relaxed for the verbal factor to achieve partial invariance (M4d); those of Arithmetic and Coding had to be freed for the Working Memory and Perceptual factors, respectively. Unfortunately, without invariance of the first-order factors, and lacking a sufficient number of invariant subtests for each factor, the latent mean differences could not be interpreted.

4. DISCUSSION

Ever since intelligence tests have been developed, the issue of gender differences in cognitive abilities aroused considerable interest and scientific debate. In the present study, we found that men performed significantly better than women in some subtests of the WAIS-IV and most of the WAIS-R. Although gender differences appeared more pronounced and pervasive in the WAIS-R sample, these results are not surprising in the light of previous research using the WAIS-III (e.g., Dolan et al., 2006; Van der Sluis et al., 2006). Examining gender differences using subtests and derived indices could be misleading: indeed most studies reported higher scores for men in verbal, perceptual, memory abilities, as well as a female advantage in the speed of cognitive processing (Daseking et al., 2017; Irwing, 2012; Longman et al., 2007).

The main issue that we addressed in the present study was to examine whether genuine differences in the latent ability factors could explain gender differences in subtests and indices. The hypothesis was not supported for Arithmetic, Information, and Comprehension, which yielded an effect size too large to be due to a difference in the underlying verbal ability or working memory. For Information and Comprehension subtests, this conclusion is reinforced by the lack of a

Table 6
Tests of measurement invariance by gender for WAIS-R and WAIS-IV using common subtests: (a) unweighted and (b) census-weighted standardization data

	(a) Unweighted data								(b) Weighted data							
WAIS-R	M1	M2	M3	M4	M4a	M4b	M4c	M4d	M1	M2	M3	M4	M4a	M4b	M4c	M4d
$\chi2$(df)	292.13	305.66	327.79	641.30	476.74	405.62	376.64	333.21	317.55	321.00	.333.80	670.81	528.50	407.67	370.98	341.64
df	54	56	65	71	70	69	68	67	54	56	65	71	70	69	68	67
CFI	.976	.975	.974	.943	.959	.966	.969	.973	.974	.973	.973	.940	.954	.966	.970	.972
NCI	.958	.956	.954	.903	.930	.942	.946	.954	.954	.954	.954	.898	.921	.941	.947	.952
TLI	.968	.968	.971	.942	.958	.965	.967	.971	.965	.966	.970	.939	.953	.964	.968	.970
RMSEA	.056	.056	.054	.076	.064	.059	.057	.053	.059	.058	.054	.078	.068	.059	.056	.054
90%CI	.050-.063	.050-.063	.048-.060	.070-.081	.059-.070	.054-.065	.051-.063	.048-.059	.053-.065	.052-.064	.049-.060	.072-.083	.063-.074	.054-.065	.051-.062	.049-.060
SRMR	.027	.031	.033	.047	.042	.038	.037	.033	.030	.031	.033	.048	.044	.037	.035	.033
$\Delta\chi2$	3.51	13.53	22.13	313.52	148.95	77.83	48.85	5.42	7.32	2.08	7.03	285.81	139.78	46.75	22.17	4.71
df($\Delta\chi2$)	6	2	9	6	5	4	3	2	6	2	9	6	5	4	3	2
p($\Delta\chi2$)	.742	.001	.008	< .001	< .001	< .001	< .001	.066	.292	.353	.635	< .001	< .001	< .001	< .001	.095
ΔCFI	.000	.001	.001	.031	.014	.007	.005	.000	.001	.000	.000	.033	.019	.007	.003	.001
ΔNCI	.000	.002	.002	.051	.024	.013	.008	.001	.000	.000	.001	.055	.032	.012	.006	.001

	(a) Unweighted data								(b) Weighted data							
WAIS-IV	M1	M2	M3	M4	M4a	M4b	M4c	M4d	M1	M2	M3	M4	M4a	M4b	M4c	M4d
$\chi2$(df)	179.27	182.66	197.44	418.21	319.95	287.27	248.71	204.66	184.54	189.39	200.69	398.67	314.99	280.51	252.33	211.39
df	54	56	65	71	70	69	68	67	54	.56	65	71	70	69	68	67
CFI	.985	.985	.984	.959	.970	.974	.979	.984	.985	.985	.984	.962	.972	.976	.979	.983
NCI	.972	.971	.970	.923	.944	.951	.959	.969	.970	.970	.970	.927	.945	.953	.958	.967
TLI	.980	.981	.983	.958	.970	.973	.977	.982	.980	.980	.983	.962	.917	.975	.978	.982
RMSEA	.046	.046	.043	.067	.057	.054	.049	.043	.047	.047	.044	.065	.057	.053	.050	.045
90%CI	.039-.054	.038-.053	.036-.050	.061-.073	.051-.064	.048-.060	.043-.056	.037-.050	.040-0.55	.040-.054	.037-.051	.059-.071	.050-.063	.047-.060	.043-.057	.038-.051
SRMR	.025	.027	.029	.042	.038	.037	.034	.030	.027	.029	.030	.042	.038	.037	.035	.032
$\Delta\chi2$	10.91	3.39	14.78	220.77	122.50	89.83	51.27	7.22	12.53	3.79	7.39	208.57	108.47	74.15	47.88	8.90
df($\Delta\chi2$)	6	2	9	6	5	4	3	2	6	2	9	6	5	4	3	2
p($\Delta\chi2$)	.091	.183	.097	< .001	< .001	< .001	< .001	.027	.051	.150	.596	< .001	< .001	< .001	< .001	.012
ΔCFI	.001	.000	.001	.025	.014	.010	.006	.001	.001	.000	.000	.022	.013	.009	.006	.001
ΔNCI	.001	.000	.001	.047	.026	.019	.011	.001	.002	.001	.001	.042	.024	.017	.011	.002

df = degrees of freedom; $\Delta\chi2$ = chi square difference test; CFI = Comparative Fit Index; ΔCFI = difference in CFI from the logically antecedent model; NCI = McDonald Non-Centrality Index; ΔNCI = difference in NCI from the logically antecedent model; TLI = Tucker-Lewin Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual.
Notes: Statistics for Configural Model (M0) used for comparison are the following: $\chi2$ = 288.62, df = 48, CFI = .976, NCI = .958 for WAIS-R unweighted; $\chi2$ = 305.17, df = 48, CFI = .974, NCI = .955 for WAIS-R weighted; $\chi2$ = 168.37, df = 48, CFI = .986, NCI = .958 for WAIS-IV unweighted; $\chi2$ = 169.01, df = 48, CFI = .986, NCI = .955 for WAIS-IV weighted. Models tested: M1 = Metric invariance FOF; M2. = Metric invariance SOF; M3 = Residual invariance. M4 = Scalar invariance FOF; M4a = Scalar invariance FOF (Information intercept freed for both WAIS-R and WAIS-IV); M4b. Scalar invariance FOF (Information and Coding intercepts freed for WAIS-R; Information and Comprehension intercepts freed for WAIS-IV); M4c. Scalar invariance FOF (Information, Coding, and Comprehension intercepts freed for WAIS-R; Information, Comprehension, and Coding intercepts freed for WAIS-IV); M4d. Scalar invariance FOF (Information, Coding, Comprehension, and Arithmetic intercepts freed for WAIS-R; Information, Comprehension, Coding, and Arithmetic intercepts freed for WAIS-IV).

latent mean difference in the verbal comprehension ability factor in the partial invariance model. Therefore, the observed gender difference at the subtest level could be due to construct-irrelevant components. For Arithmetic, although there was a latent mean difference in the working memory ability factor, subsequent analyses illustrated that the actual mean difference in the subtest score was much above that one can expect considering only the difference in the corresponding ability.

Regarding Information, studies using older editions of the Wechsler scales for adults showed a gender difference in this subtest (Arceneaux et al., 1996; Lynn et al., 2005; Lynn & Dai, 1993; Snow & Weinstock, 1990). This finding was replicated in subsequent analyses of the WAIS-III (Irwing, 2012; Van der Sluis et al., 2006). Our study confirmed the male advantage in this specific subtest using the WAIS-IV. Moreover, it suggested that there might be something specific about Information that cannot be explained fully by cognitive ability factors. We believe that, although the Information subtest should measure general knowledge gathered from everyday situations, the item-content might fit more with the typical knowledge domains that appeal to a male audience. For instance, the literature has shown that geographic or scientific information belongs to stereotypical male professions or informal educational routes (e.g., Dabbs Jr., Chang, Strong, & Milun, 1998; Lawton, 2018). Already in a paper by Ilai and Willerman (1989) emerged that a significant proportion of the items in the Information subtest of WAIS-R were more suitable for men than for women.

The Arithmetic subtest includes items in which people have to solve mathematical problems presented verbally with time limits and without the support of paper and pencil. Our study showed that the effect size for this subtest was the largest among all subtests and across editions of the Wechsler scales. Previous research has repeatedly found a male advantage in arithmetic tasks (e.g., Arceneaux et al., 1996; Held et al., 1993; Irwing, 2012; Snow & Weinstock, 1990), although the gender gap seems narrowing (Guiso, Monte, Sapienza, & Zingales, 2008; Hyde, Lindberg, Linn, Ellis, & Williams, 2008; Hyde & Mertz, 2009). Our study suggests that gender differences in Arithmetic could be a significant source of concern if this subtest entails a severe test bias in favor of men. Similar to the Information subtests, mathematics is stereotypically a matter in which men are more encouraged than women (Reuben, Sapienza, & Zingales, 2014). Nevertheless, an alternative explanation could be that of minor differences in working memory that need further consideration in light of the latent mean difference in the working memory ability factor. We believe future research is needed to address this issue using "pure" measures of working memory ability that not include arithmetic calculation. Indeed, inferences based on a latent mean difference should be avoided if one or more indicators of the factor are not scalar invariant, as it seems to be the case.

As in a recent study of the German WAIS-IV (Daseking et al., 2017), we found a small male advantage in the Comprehension subtest that appeared not entirely justified examining its collinearity with the factor loading. The Comprehension subtest entails knowledge of social conventions along with verbal expression and reasoning abilities. Since we have not found explanations in the literature on reasons for these gender differences, we speculated that some social situations or general concepts described by the items could be less familiar to women than to men. For instance, some items ask about political, economic, and legal issues that are not topics routinely discussed by women in everyday social situations (e.g., international relationships between countries; the cost of the land in the country and the city).

The multigroup analysis of the WAIS-IV showed that the full scalar invariance could not be achieved. In keeping, with the findings discussed above, we found that the intercepts of Information, Arithmetic, and Comprehension were not invariant by gender. This finding means that there was a systematic upward bias in these subtests that favors men. Before discussing the implications of this finding in terms of consequences on the assessment of verbal comprehension and working memory abilities, it worth noting that neither the perceptual reasoning subtests nor the processing speed ones were biased. We could trust both the effect size differences for the corresponding indices in Table 2 and the latent mean differences in the multigroup analysis. Indeed, women and men did not differ in the processing speed ability, while the better performance of men in perceptual reasoning was small difference according to Hyde's guidelines.

The finding that men performed better than women in tasks involving perceptual reasoning abilities is well-established in previous research using the WAIS-III (Dolan et al., 2006; Van der Sluis et al., 2006). A recent study using the WAIS-IV has also shown a higher score in PRI for men (Daseking et al., 2017). Our findings add to the extant literature, confirming that gender differences in PRI and the corresponding subtests depend on the latent ability factor, although this difference likely is a small effect size.

Regarding processing speed abilities, previous research provided mixed results. For instance, Dutch and American studies of the WAIS-III (Irwing, 2012; Van der Sluis et al., 2006) and the German study of the WAIS-IV (Daseking et al., 2017) showed that women were higher, and indeed faster, than men on the latent mean of the processing speed factor and the PSI index, respectively. By contrast, the Spanish study of the WAIS-III did not detect any gender difference, as in our study. Processing speed has been frequently linked with biological components of intelligence (e.g., Majeres, 2007; Pugh et al., 1997); however, scholars also maintain that "tests of processing speed are probably not more culture-free than other cognitive tests and thus they are not a reliable instrument in assessing cross-national or cross-ethnic differences in intelligence" (Roivainen, 2011, p. 149).

The multigroup analysis of the WAIS-R also showed that the full scalar invariance could not be met. However, the subtests whose intercepts had to be relaxed to achieve the partial measurement invariance were Arithmetic, Information, Coding, Digit Span, and Comprehension. These results impeded the assessment of gender differences at the latent mean difference level. All the verbal subtests (including Arithmetic and Digit-Span, which are today considered proxy measures of working memory abilities) showed upward biases that favor men. Instead, the Coding subtest (today considered a marker of processing speed) showed an upward bias that favors women. Noteworthy, the intercepts of the performance subtests today used to score the PRI (i.e., Block Design and Picture Completion) were gender invariant.

Comparing the bias accounted for by intercept differences in Information, Comprehension, and Arithmetic resulting from the WAIS-R normative sample to corresponding estimates obtained from the WAIS-IV, we noted that the bias in Information and Comprehension was about the same across the two editions. Instead, the bias in the Arithmetic subtest is reduced from +19% to +11% in favor of men. Because the two normative samples were collected spaced 16 years apart, gender differences in Arithmetic may be becoming less important today than in the past. Since Terman (1916) and Wechsler (1944), it appeared that men performed better than women in complex numerical tasks. Research contemporary to the normative sample of the WAIS-R used in this study has also shown gender differences in Arithmetic of the Wechsler scales and other cognitive tests (e.g., Arceneaux et al., 1996; Held et al., 1993; Snow & Weinstock, 1990). Similarly, the research conducted in the second half of the past century had highlighted the gender differences in math performance (Hyde, Fennema, & Lamon, 1990). However, more recent studies suggest that the gender gap in math performance is shrinking (Hyde et al., 2008; Hyde & Mertz, 2009) and so could be narrowing the bias for the Arithmetic subtest in the WAIS-IV normative sample.

The intercept of Digit Span was also non-invariant by gender in the WAIS-R, with a male advantage of +10% that disappeared in the WAIS-IV. It should be kept in mind that the Arithmetic and Digit Span subtests are the most different in item composition in the two editions of the WAIS. For example, the Arithmetic subtest includes new problems, and the Digit-Span subtest is now comprised of three parts (i.e., Forward, Backward, and Sequencing). More generally, the WAIS-IV was

developed to be more theoretically close to the current models of abilities (e.g., Benson et al., 2010), moving away for the simplistic view of verbal and nonverbal IQ. Indeed, former verbal subtests of the WAIS-R, like Arithmetic and Digit Span, are no longer considered verbal subtests in the WAIS-IV, forming a separate Working Memory Index (WMI). We sought to compare the two MGCFA using the common subtests only, assuming a hierarchical structure with three first-order factors that proved useful to identify a working memory-like factor different from verbal and performance factors in the Italian WAIS-R. This attempt did not improve the measurement equivalence of the WAIS-R and worsened the number of intercepts to be freed for the WAIS-IV to achieve the partial scalar invariance.

This study was not exempt from limitations. The first and foremost is related to the potential lack of representativeness for the male and female sub-samples. Both the WAIS-IV and WAIS-R normative data were non-probability samples of voluntary subjects recruited from the general population. So there could be a volunteer bias in that those with a higher interest in psychological inquiries would participate in the study. Also, respondents were chosen using approximate equal quotas for gender but not for age and education. Whereas the overall datasets included as many women as men, we noted that education and age were distributed unequally within each gender group. Since we could not change the composition of the two normative samples, we have committed ourselves to assess how the joint distribution of gender, age, and education deviated from Italian census data. The best we could do, under the circumstances, was developing sample weights to match the demographics of the Italian population in census-weighted analyses. This solution does not entirely fix the sampling and volunteer biases, and then caution is needed in generalizing the current findings to whole male and female populations.

A second noteworthy limitation of the present study was the inability to compare the latent means of the three ability factors in the analysis of WAIS-IV and WAIS-R using the common subtests. The modification indices suggested that the intercepts of four subtests out of nine had to be relaxed to achieve the partial scalar invariance. Given the relatively small number of common subtests, there was an insufficient number of invariant indicators to enable an unbiased comparison of latent mean differences. This finding tempered our hope to take advantage of datasets spaced 16 years apart to observe generational differences associated with gender differences in broad cognitive abilities — nevertheless, these analyses allowed us to examine generational differences in the intercepts of specific subtests.

Third, all the analyses used age-standardized subtest scores. A possible inconvenience of this choice could be the reduced heterogeneity of the subtests and derived indices, with particular reference to male variability in IQ that has been documented as a presumed cause of gender differences in intelligence (e.g., Dykiert et al., 2009). The use of age-standardized scores has, however, made it possible to monitor for any spurious effect of age in the context of a non-probability sample that was our primary concern.

Notwithstanding these limitations, our study has shown that the three subtests of the WAIS-IV should be looked at carefully because they may conceal a potential measurement bias in the assessment of verbal ability and working memory. Gender differences in perceptual reasoning seem genuine instead, as does their absence in processing speed. Further evidence emerging from this study is the reduction of the men-women gap in non-cognitive factors related to performance in the arithmetic subtest in two editions of the WAIS. Finally, it should be noted that despite this study adds to a wide range of research showing that the Information subtest undoubtedly affects the assessment of gender differences, it continues to be used as a core subtest for the evaluation of verbal ability.

Before concluding, it is worth discussing some implications for psychological assessment. Assuming the remaining verbal subtests to be not biased and calculating the VCI using the weighted sum of the core subtests only, the observed average gender difference in the VC score would be +0.67 in favor of men. According to the WAIS-IV conversion table (VC ➔ VCI), this difference would correspond to +2 IQ points. Correcting this estimate for test bias due to the non-invariant intercept of Information, the average gender difference would be in favor of women corresponding to -0.46 VC, which would be a null difference in terms of IQ points. The same reasoning, applied to the WMI, would lead to an observed average gender difference in WM equal to +1.90 in favor of men, which would correspond to +6 IQ points, using bias-corrected data, this difference would be +1.05, that means +3 IQ points only. It is worth reminding that a bias of 2-3 IQ points corresponds to about one-fifth of standard deviation, which is perhaps not very influential in assessment situations using the composite indices or full-scale IQ. Greater caution should be exercised in the interpretation of subtest scaled scores profile in the light of the gender differences in the intercepts of the three non-invariant subtests.

To conclude, our study highlights that the WAIS-IV was quite fair in the evaluation of gender differences, which, although significant at the intercept level, were less pronounced than those resulting from the WAIS-R.

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REFERENCES

Aluja-Fabregat, A., Colom, R., Abad, F., & Juan-Espinosa, M. (2000). Sex differences in general intelligence defined as g among young adolescents. *Personality and Individual Differences, 28*, 813–820.

Ankney, C. D. (1992). Sex differences in relative brain size: The mismeasure of woman, too? *Intelligence, 16*(3-4), 329–336.

Arceneaux, J. M., Cheramie, G. M., & Smith, C. W. (1996). Gender differences in WAIS-R age-corrected scaled scores. *Perceptual and Motor Skills, 83*(3, Pt 2), 1211–1215. https://doi.org/10.2466/pms.1996.83.3f.1211.

Benson, N., Hulac, D. M., & Kranzler, J. H. (2010). Independent examination of the Wechsler Adult Intelligence Scale—Fourth Edition (WAIS-IV): what does the WAIS-IV measure? *Psychological Assessment, 22*(1), 121.

Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal, 14*(3), 464–504.

Colom, R., García, L. F., Juan-Espinosa, M., & Abad, F. J. (2002). Null sex differences in general intelligence: Evidence from the WAIS-III. *The Spanish Journal of Psychology, 5*, 29–35.

Colom, R., & Garcia-Lopez, O. (2002). Sex differences in fluid intelligence among high school graduates. *Personality and Individual Differences, 32*, 445–451.

Colom, R., Juan-Espinosa, M., Abad, F., & García, L. F. (2000). Negligible sex differences in general intelligence. *Intelligence, 28*(1), 57–68.

Dabbs, J. M., Jr., Chang, E.-L., Strong, R. A., & Milun, R. (1998). Spatial ability, navigation strategy, and geographic knowledge among men and women. *Evolution and Human Behavior, 19*(2), 89–98. https://doi.org/10.1016/S1090-5138(97)00107-4.

Daseking, M., Petermann, F., & Waldmann, H. C. (2017). Sex differences in cognitive abilities: Analyses for the German WAIS-IV. *Personality and Individual Differences, 114*, 145–150.

De Beuckelaer, A., & Swinnen, G. (2018). Biased Latent Variable Mean Comparisons due to Measurement Noninvariance: A Simulation Study. In E. Davidov, P. Schmidt, J. Billiet, & B. Meuleman (Eds.). *Cross-cultural analysis* (pp. 127–156). Routledge.

Dolan, C. V., Colom, R., Abad, F. J., Wicherts, J. M., Hessen, D. J., & van de Sluis, S. (2006). Multi-group covariance and mean structure modeling of the relationship between the WAIS-III common factors and sex and educational attainment in Spain. *Intelligence, 34*, 193–210.

Dykiert, D., Gale, C. R., & Deary, I. J. (2009). Are apparent sex differences in mean IQ scores created in part by sample restriction and increased male variance? *Intelligence, 37*, 42–47.
Guiso, L, Monte, F, Sapienza, P, & Zingales, L (2008). Culture, math, and gender. *Science, 320*(5880), 1164–1165.
Held, J. D., Alderton, D. L., Foley, P. P., & Segall, D. O. (1993). Arithmetic reasoning gender differences: explanations found in the Armed Services Vocational Aptitude Battery (ASVAB). *Learning and individual differences, 5*(2), 171–186.
Hunt, E., & Madhyastha, T. (2008). Recruitment modeling: An analysis and an application to the study of male–female differences in intelligence. *Intelligence, 36*(6), 653–663.
Hyde, J. S. (2005). The gender similarities hypothesis. *American psychologist, 60*(6), 581.
Hyde, J. S. (2016). Sex and cognition: gender and cognitive functions. *Current opinion in neurobiology, 38*, 53–56.
Hyde, J. S., Fennema, E., & Lamon, S. J. (1990). Gender differences in mathematics performance: A meta-analysis. *Psychological bulletin, 107*(2), 139.
Hyde, JS, Lindberg, SM, Linn, MC, Ellis, AB, & Williams, CC (2008). Diversity. Gender similarities characterize math performance. *Science, 321*(5888), 494–495.
Hyde, JS, & Mertz, JE (2009). Gender, culture, and mathematics performance. *Proceedings of the National Academy of Sciences of the United States of America, 106*(22), 8801–8807.
Ilai, D., & Willerman, L. (1989). Sex differences in WAIS-R item performance. *Intelligence, 13*(3), 225–234. https://doi.org/10.1016/0160-2896(89)90019-6.
Irwing, P. (2012). Sex differences in g: An analysis of the US standardization sample of the WAIS-III. *Personality and Individual Differences, 53*, 126–131.
Irwing, P., & Lynn, R. (2005). Sex differences in means and variability on the progressive matrices in university students: A meta-analysis. *British Journal of Psychology, 96*, 505–524.
Jackson, D. N., & Rushton, J. P. (2006). Males have greater g: Sex differences in general mental ability from 100,000 17- to 18-year-olds on the Scholastic Assessment Test. *Intelligence, 34*, 479–486.
Kang, Y., McNeish, D. M., & Hancock, G. R. (2016). The role of measurement quality on practical guidelines for assessing measurement and structural invariance. *Educational and psychological measurement, 76*(4), 533–561.
Keith, T. Z., Reynolds, M. R., Patel, P. G., & Ridley, K. P. (2008). Sex differences in latent cognitive abilities ages 6 to 59: Evidence from the Woodcock Johnson III tests of cognitive abilities. *Intelligence, 36*, 502–525.
Kuha, J., & Moustaki, I. (2015). Nonequivalence of measurement in latent variable modeling of multigroup data: A sensitivity analysis. *Psychological methods, 20*(4), 523.
Lawton, C. A. (2018). Sex and gender in geographic behavior and cognition. In D. R. Montello (Ed.). *Handbook of behavioral and cognitive geography* (pp. 247–259). Northampton, MA: Edward Elgar Publishing. https://doi.org/10.4337/9781784717544.00023.
Longman, R. S., Saklofske, D. H., & Fung, T. S. (2007). WAIS-III percentile scores by education and sex for US and Canadian populations. *Assessment, 14*(4), 426–432.
Lynn, R. (1994). Sex differences in brain size and intelligence: A paradox resolved. *Personality and Individual Differences, 17*, 257–271.
Lynn, R. (1998). Sex differences in intelligence: Data from a Scottish standardisation of the WAIS-R. *Personality and Individual Differences, 24*(2), 289–290.
Lynn, R. (1999). Sex differences in intelligence and brain size: A developmental theory. *Intelligence, 27*, 1–12.
Lynn, R., & Dai, X.-Y. (1993). Sex differences on the Chinese standardization sample of WAIS-R. *The Journal of Genetic Psychology: Research and Theory on Human Development, 154*(4), 459–463. https://doi.org/10.1080/00221325.1993.9914744.
Lynn, R., Fergusson, D. M., & Horwood, L. J. (2005). Sex differences on the WISC-R in New Zealand. *Personality and Individual Differences, 39*(1), 103–114. https://doi.org/10.1016/j.paid.2004.12.009.
Lynn, R., & Irwing, P. (2004a). Sex differences on the advanced progressive matrices in college students. *Personality and Individual Differences, 37*, 219–223.
Lynn, R., & Irwing, P. (2004b). Sex differences on the progressive matrices: A meta-analysis. *Intelligence, 32*, 481–498.
Lynn, R., & Irwing, P. (2008). Sex differences in mental arithmetic, digit span, and g defined as working memory capacity. *Intelligence, 36*, 226–235.
Majeres, R. L. (2007). Sex differences in phonological coding: Alphabet transformation speed. *Intelligence, 35*, 335–346.
McDaniel, M. A. (2005). Big-brained people are smarter: A meta-analysis of the relationship between in vivo brain volume and intelligence. *Intelligence, 33*(4), 337–346.
Meade, A. W., Johnson, E. C., & Braddy, P. W. (2008). Power and sensitivity of alternative fit indices in tests of measurement invariance. *Journal of applied psychology, 93*(3), 568.
Nyborg, H. (2003). Sex differences in g. In H. Nyborg (Ed.). *The scientific study of general intelligence. Tribute to Arthur R. Jensen* (pp. 187–222). Amsterdam: Elsevier Science Ltd.
Nyborg, H. (2005). Sex-related differences in general intelligence g, brain size, and social status. *Personality and Individual Differences, 39*, 497–509.
Oberski, D. L. (2014). lavaan.survey: an R package for complex survey analysis of structural equation models. *Journal of Statistical Software, 57*(1), 1–27. https://doi.org/10.18637/jss.v057.i01.
Orsini, A., & Laicardi, C. (1997). *WAIS-R. Contributo alla taratura italiana.* Firenze: O.S. Organizzazioni Speciali.
Orsini, A., & Laicardi, C. (2003). *WAIS-R e terza età. La natura dell'intelligenza nell'anziano: continuità e discontinuità.* Firenze: O.S. Organizzazioni Speciali (seconda ristampa, 2009).
Orsini, A., & Pezzuti, L. (2013). *WAIS-4: contributo alla taratura italiana (16-69 anni).* Giunti OS.
Orsini, A., & Pezzuti, L. (2015). *WAIS-4: contributo alla taratura italiana (70-90 anni).* Giunti OS.
Parro, F. (2012). International evidence on the gender gap in education over the past six decades: A puzzle and an answer to it. *Journal of Human Capital, 6*, 150–185. https://doi.org/10.1086/666849.
Pezzuti, L., & Orsini, A. (2016). Are there sex differences in the Wechsler Intelligence Scale for Children – Forth Edition? *Learning and Individual Differences, 45*, 307–312.
Pezzuti, L., & Rossetti, S. (2017a). Number Sequencing, Figure Weights, and Cancellation subtests of WAIS-IV administered to elders. *Personality and Individual Differences, 104*, 352–356.
Pezzuti, L., & Rossetti, S. (2017b). Norms for Letter and Number Sequencing, Figure Weights and Cancellation subtests for the elderly Italian population. *BPA-Applied Psychology Bulletin (Bollettino di Psicologia Applicata), 65*(279).
Pugh, K. R., Shaywitz, B. A., Shaywitz, S. E., Shankweiler, D. P., Katz, L., Fletcher, J., et al. (1997). Predicting reading performance from neuroimaging profiles: The cerebral basis of phonological effects in printed word identification. *Journal of Experimental Psychology: Human Perception and Performance, 23*, 145–151.
Reuben, E., Sapienza, P., & Zingales, L. (2014). How stereotypes impair women's careers in science. *Proceedings of the National Academy of Sciences, 111*(12), 4403–4408.
Reynolds, M. R., Keith, T. Z., Ridley, K. P., & Patel, P. G. (2008). Sex differences in latent general and broad cognitive abilities for children and youth: Evidence from higher-order MG-MACS and MIMIC models. *Intelligence, 36*, 236–260.
Rindermann, H. (2007). The big g-factor of national cognitive ability. *European Journal of Personality, 21*, 767–787.
Roivainen, E. (2011). Gender differences in processing speed: A review of recent research. *Learning and Individual differences, 21*(2), 145–149.
Rosseel, Y. (2012). lavaan: An R Package for Structural Equation modeling. *Journal of Statistical Software, 48*(2), 1–36.
Rudnev, M., Lytkina, E., Davidov, E., Schmidt, P., & Zick, A. (2018). Testing measurement invariance for a second-order factor: a cross-national test of the Alienation Scale. *Methods, Data, Analyses: A Journal for Quantitative Methods and Survey Methodology (MDA), 12*(1), 47–76.
Rushton, J. P., & Ankney, C. D. (2009). Whole brain size and general mental ability: a review. *International Journal of Neuroscience, 119*(5), 692–732.
Saggino, A., Pezzuti, L., Tommasi, M., Cianci, L., Colom, R., & Orsini, A. (2014). Null sex differences in general intelligence among elderly. *Personality and Individual Differences, 63*, 53–57.
Schermelleh-Engel, K., Moosbrugger, H., & Müller, H. (2003). Evaluating the Fit of Structural Equation Models: Tests of Significance and Descriptive Goodness-of-Fit Measures. *Methods of Psychological Research Online, 8*, 23–74.
Snow, W. G., & Weinstock, J. (1990). Sex differences among non-brain-damaged adults on the Wechsler Adult Intelligence Scales: A review of the literature. *Journal of Clinical and Experimental Neuropsychology, 12*(6), 873–886. https://doi.org/10.1080/01688639008401028.
Steinmayr, R., Beauducel, A., & Spinath, B. (2010). Do sex differences in a faceted model of fluid and crystallized intelligence depend on the method applied? *Intelligence, 38*(1), 101–110.
Terman, L. M. (1916). *The measurement of intelligence.* Boston, MA: Houghton Mifflin.
Tommasi, M., Watkins, M., Orsini, A., Pezzuti, L., Cianci, L., & Saggino, A. (2015). Gender differences in latent cognitive abilities and education links with g in Italian elders. *Learning and Individual Differences, 37*, 276–282.
Van der Sluis, S., Derom, C., Thiery, E., Bartels, M., Polderman, T. J. C., Verhulst, F. C., et al. (2008). Sex differences on the WISC-R in Belgium and the Netherlands. *Intelligence, 36*, 48–67.
Van der Sluis, S., Posthuma, D., Dolan, C. V., de Geus, E. J. C., Colom, R., & Boomsma, D. I. (2006). Sex differences on the Dutch WAIS-III. *Intelligence, 34*, 273–289.
Wechsler, D. (1944). *Measurement of adult intelligence* (3rd ed). Baltimore: William and Wilkins.
Wechsler, D. (1981). *Wechsler Adult Intelligence Scale – Revised.* New York: The Psychological Corporation.
Wechsler, D. (2008). *Wechsler adult intelligence scale–Fourth edition (WAIS–IV).* San Antonio, TX: NCS Pearson 22, 498.
Wicherts, J. M., & Dolan, C. V. (2010). Measurement invariance in confirmatory factor analysis: An illustration using IQ test performance of minorities. *Educational Measurement: Issues and Practice, 29*(3), 39–47.